

Analyze changes of secondary-structure ensembles during transcription over time

Overview

ff-co_timecourse simulates co-transcriptional RNA folding. The program extends the sequence nucleotide by nucleotide and performs a stochastic kinetic folding simulation for each transcript length. Macro-state occupancies are recorded over time and visualized.

Input files

The file **dld3.fa** contains a designed RNA sequence:

```
>dld3.fa
UCAGUCUUCGCGCUGCUGUAUCGAUUCGGUUUCAGUUUUUAUUGC
```

If you want to start simulations in a specific **folded structure**, provide that structure explicitly. Through this structure you can also give the transcript length you want the simulation to start from. When no initial structure is given, the simulations start from transcript length 1. Here is an example with a starting length of three from an unfolded structure:

```
>dld3
UCAGUCUUCGCGCUGCUGUAUCGAUUCGGUUUCAGUUUUUAUUGC
...
```

Macro-states

To partition the overall secondary-structure ensemble into smaller ensembles of interest, we define **macro-states** using files such as **dld3_lm*.ms**.

Example (**dld3_lm3_3.0.ms**):

```
>LM3 lmin=lm3_bh=3.0
UCAGUCUUCGCGCUGCUGUAUCGAUUCGGUUUCAGUUUUUAUUGC
.((((.....))).((((.....))).
.((((.....))).((((.....))).
.((((.....))).((((.....))).
.((((.....))).((((.....))).
.((((.....))).((((.....))).
.((((.....))).((((.....))).
.((((.....))).((((.....))).
.((((.....))).((((.....))).
.((((.....))).((((.....))).
.((((.....))).((((.....))).
```

Here:

- The first line defines the **macro-state name** (`LM3`) and, optionally, some more description after a white-space (`lmin=lm3_bh=3.0`).
 - The second line specifies the **sequence**.
 - The remaining lines list all **secondary structures** that belong to this macro-state.
-

Input Parameters

You can specify the transcription rate via the extension time (`--t-ext`), which defines the simulation time per transcript length. The posttranscriptional folding time (`--t-end`) defines the simulation time at full sequence length. To express simulation time in seconds, set the rate constant `k0` to `1e6`. Furthermore, you can specify timeline parameters:

- `--t-lin`: number of linearly spaced time points recorded per transcript length
 - `--t-log`: number of logarithmically spaced time points recorded during posttranscriptional folding.
- To further familiarize yourself with the parameters:

```
ff-co_timecourse --help
```

Simulation setup

Note: Ensure the `fuzzyfold` package is installed.

When working directly from the Git repository, use

```
cargo run --bin ff-co_timecourse -- [options]
```

instead of calling `ff-co_timecourse` directly.

To simulate 100 trajectories starting at transcript length 1:

```
cat dld3.fa | ff-co_timecourse --macrostates dld3*.ms --k0 1e6 --t-lin 10  
--t-log 100 --t-ext 0.02 --t-end 1.0 -n 100
```

or equivalently:

```
ff-co_timecourse --macrostates dld3*.ms --k0 1e6 --t-lin 10 --t-log 100 --  
t-ext 0.02 --t-end 1.0 -n 100 < dld3.fa
```

During execution, the program prints simulation parameters to `STDOUT`, displays a **progress bar**, and outputs **time-course data** once all runs are completed. The time course is also plotted automatically as an SVG file, where the plot name is derived from the input file. For example:

```
ff_dld3.svg
```

Pausing sites

When you do not want an uniform extension time, but define pausing sites, you can define the positions and the duration of the pauses in two separate input parameters: **p-pos** are the positions of the pausing sites and **t-pos** the respective duration. To have a pause at position 2 for 1s and one at position 10 for 0.5s:

```
cat dld3.fa | ff-co_timecourse --macrostates dld3*.ms --k0 1e6 --t-lin 10
--t-log 100 --t-ext 0.02 --t-end 1.0 --p-pos 2,10 --t-pos 1,0.5 -n 100
```

Aggregating data from multiple simulations

To reduce statistical noise in ensemble dynamics, you may want to perform **many more trajectories**, potentially for longer time periods. You can *accumulate results incrementally* by reloading existing timelines.

For example:

```
cat dld3.fa | ff-co_timecourse --macrostates dld3*.ms --k0 1e6 --t-lin 10
--t-log 100 --t-ext 0.02 --t-end 1.0 -n 100 --timeline my_dld3.tln
```

This command creates **my_dld3.tln**, which stores the results from 100 simulations. Running the same command again will automatically reload the file, add another 100 simulations, and update the stored timeline accordingly.

Try it! This is the recommended way to extend your simulation dataset without restarting from scratch.

An example output file from 1000 aggregated simulations with the following parameters: `--k0 1e6 --t-lin 10 --t-log 100 --t-ext 0.02 --t-end 1.0` may look like this:

